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Analytical Geometry Paper 2

Several facts from Euclidean Geometry can be tested in this section.

Know :

(1) Triangles

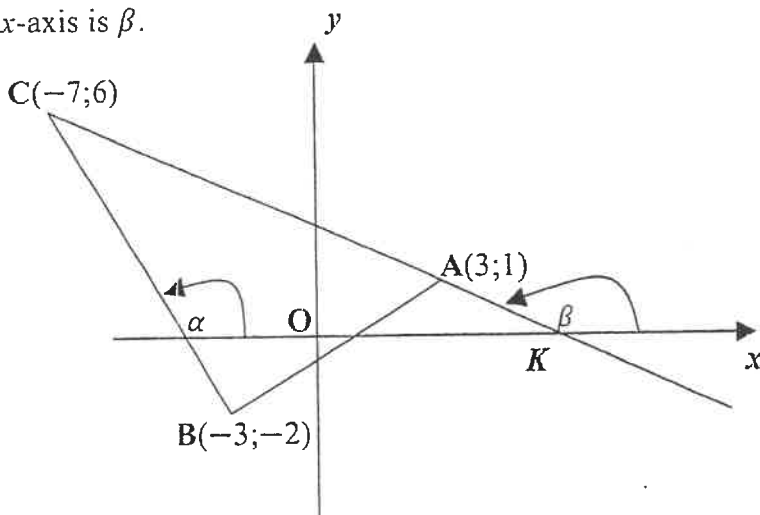
- All geometry of Triangles.
- Special lines : medians, altitudes, perpendicular bisectors.
- Area of a triangle

(2) Quadrilaterals - all properties.

(3) Circle Geometry - all theorems.

QUESTION 1

$\triangle ABC$ with $A(3;1)$, $B(-3;-2)$ and $C(-7;6)$ are given. CAK is a straight line. The angle that BC forms with the positive x -axis is α and the angle that CK forms with the positive x -axis is β .



- 1.1 Calculate the gradient of AC. (2)
- 1.2 Calculate the coordinates of K. (4)
- 1.3 Show that $AB \perp BC$. (3)
- 1.4 Calculate the area of $\triangle ABC$. Leave your answer in simplest surd form. (5)
- 1.5 If $BD \perp AC$, determine the equation of BD . (4)
- 1.6 Calculate $\widehat{BCA}$, correct to the nearest degree. (5)

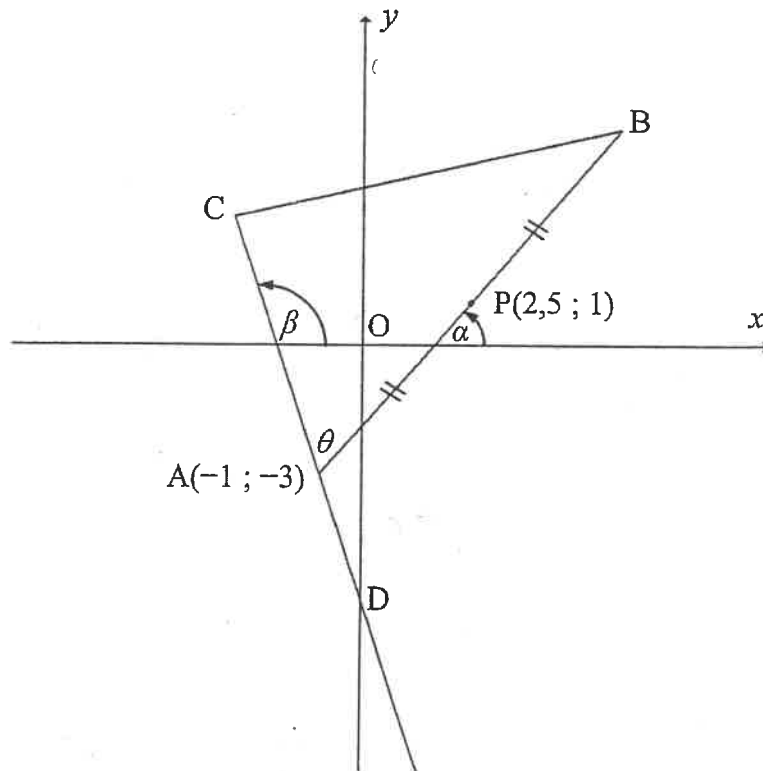
QUESTION. 2

In the diagram below, $A(-1 ; -3)$, B and C are the vertices of a triangle.

$P(2,5 ; 1)$ is the midpoint of AB. CA, extended cuts the y-axis at D.

The equation of CD is $y = -3x + k$. $\hat{CAB} = \theta$.

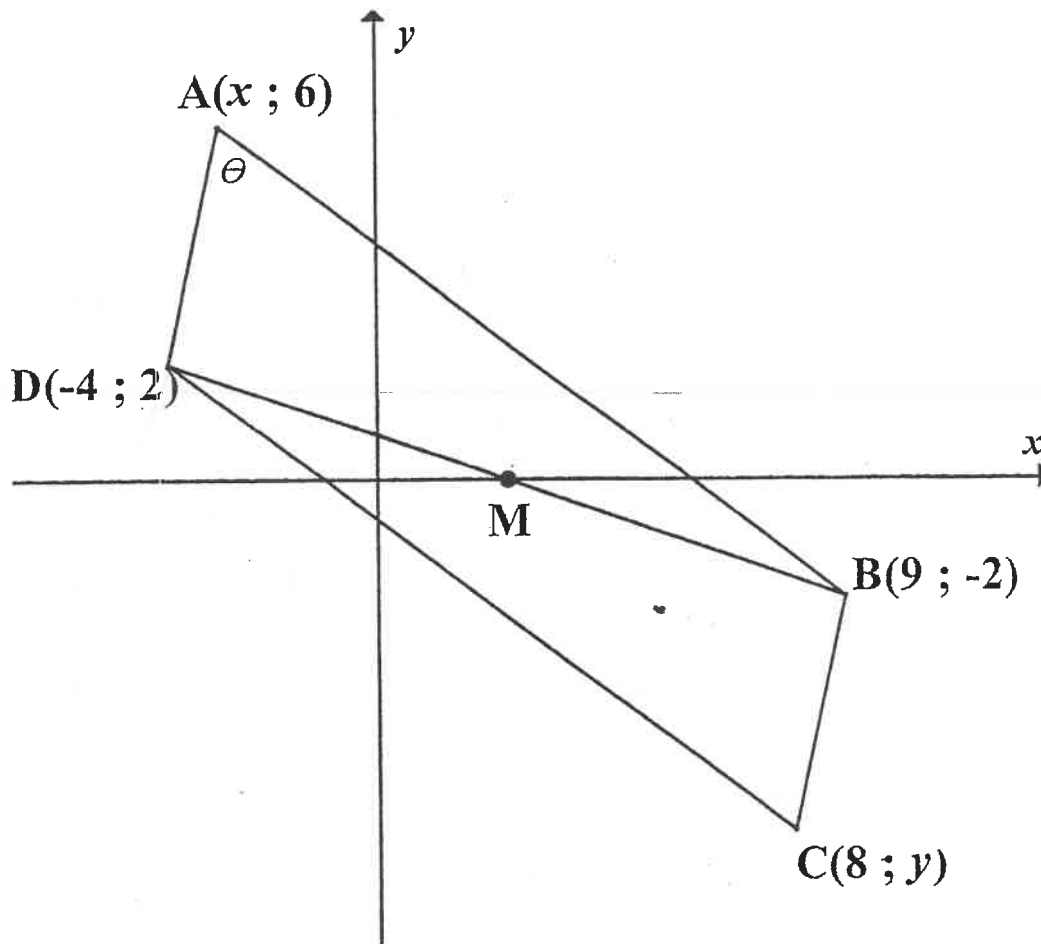
α and β are the angles that AB and AC respectively make with the x-axis.



- | | | |
|-----|---|-----|
| 2.1 | Determine the value of k . | (2) |
| 2.2 | Determine the coordinates of B. | (2) |
| 2.3 | Determine the gradient of AB. | (2) |
| 2.4 | Calculate the size of θ . | (5) |
| 2.5 | Calculate the length of AD. Leave your answer in surd form. | (2) |
| 2.6 | If $AC = 2AD$ and $AB = \sqrt{113}$, calculate the length of CB. | (5) |

QUESTION 3

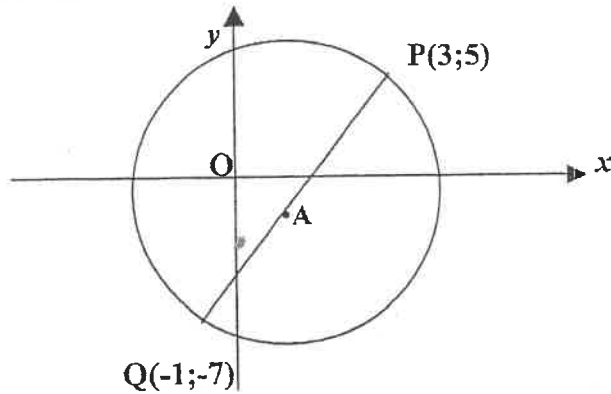
In the diagram below ABCD is a parallelogram.



- | | | |
|-----|--|-----|
| 3.1 | Calculate the coordinates of M, the midpoint of DB. | (2) |
| 3.2 | Show, by calculation that the value of $x = -3$ and $y = -6$. | (3) |
| 3.3 | Calculate the gradient of BC. | (2) |
| 3.4 | Calculate the equation of the line AB in the form $y = \dots$ | (4) |
| 3.5 | Determine θ . | (4) |
| 3.6 | Prove that $\triangle DBC$ is not a right angled triangle | (3) |

QUESTION 4

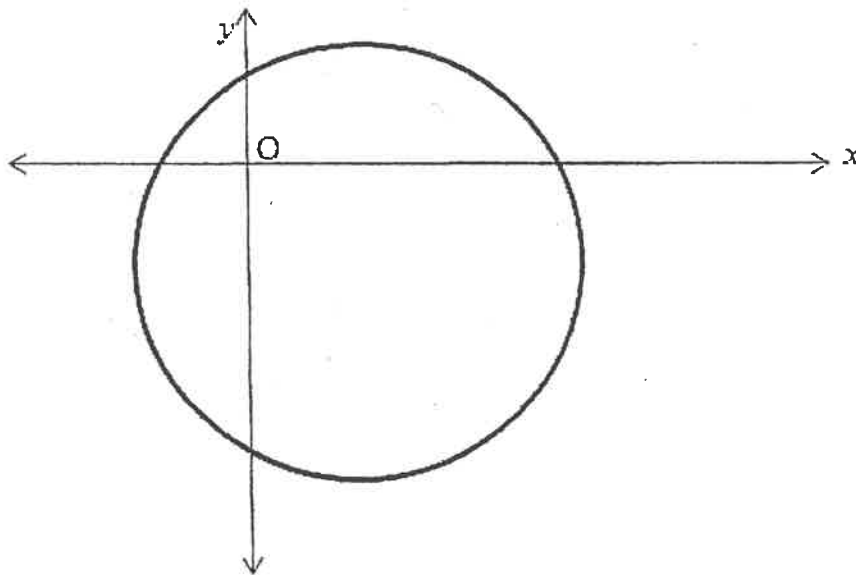
PQ is the diameter of the circle with $P(3;5)$ and $Q(-1;-7)$. A is the centre of the circle.



- 4.1. Write down the coordinates of A. (2)
- 4.2. Hence calculate the equation of the circle in the form $(x - a)^2 + (y - b)^2 = r^2$ (3)

QUESTION 5

The length of the radius of the circle with equation: $x^2 - 2x + y^2 + 4y = a$ is 5 units.



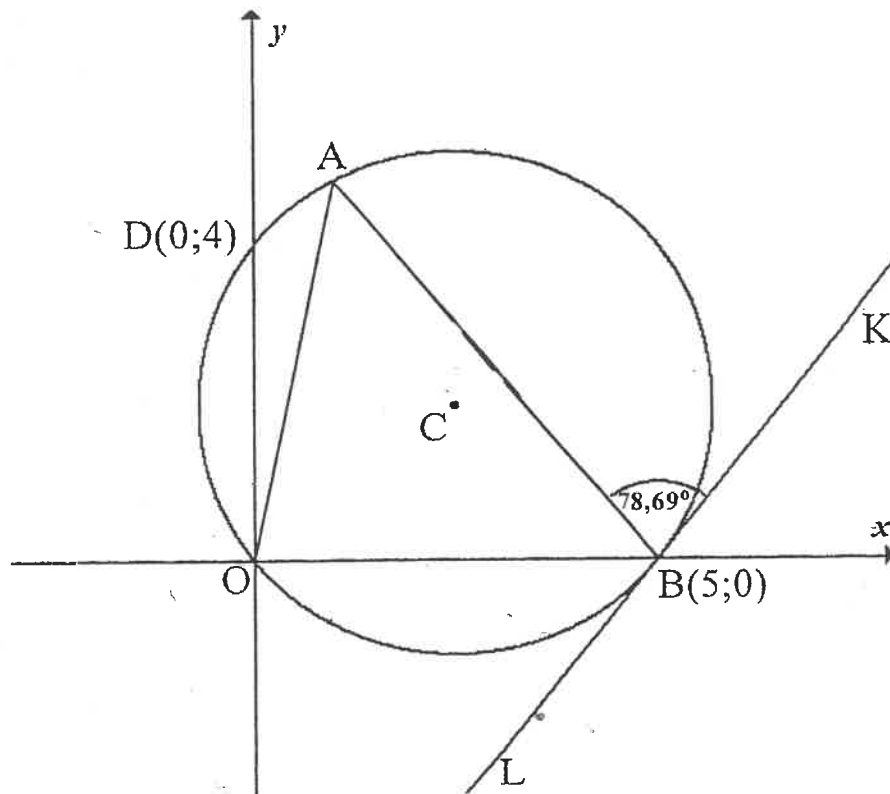
- 5.1 Show by means of calculations that $a = 20$ units. (4)
- 5.2 Write down the coordinates of the centre M of the circle. (1)
- 5.3 $A(x; y)$, with $y > 0$, is one of the points of intersection of the circle and the straight line $x = 4$. Determine the values of x and y . (4)
- 5.4 Determine the equation of the tangent to the circle at the point A. (4)
- 5.5 Determine whether the point $T(-1; -2)$ lies inside or outside the circle. (3)
- 5.6 If the circle is translated 3 units to the left and 1 unit up, determine the equation of the new circle. (3)

QUESTION 6

In the diagram below is the sketch of a circle with centre $C\left(\frac{5}{2}; 2\right)$.

The circle goes through the origin and cuts the x - and y -axis at B and D respectively.

LBK is tangent to the circle at B. $\angle ABK = 78,69^\circ$.



- 6.1 Show by calculation that the equation of the circle is $x^2 + y^2 - 5x - 4y = 0$. (4)

- 6.2 Determine the equation of LBK in the form $y = mx + c$. (5)

- 6.3 Determine the size of $\angle AOB$. (1)

- 6.4 Determine the gradient of OA correct to the nearest integer and hence the equation of OA. (2)

- 6.5 Calculate the coordinates of A. (5)

- 6.6 If there is another circle

$$\left(x + \frac{5}{2}\right)^2 + (y - 3)^2 = \frac{61}{4}$$

- 3.6.1 Show that the two circles intersect at two distinct points. (3)

Analytical Geometry: Solutions

Question 1:

$$1.1. \quad m = \frac{6-1}{-7-3}$$

$$= \underline{-\frac{1}{2}},$$

$$1.2. \quad y - y_1 = m(x - x_1)$$

$$y - 1 = -\frac{1}{2}(x - 3)$$

$$y - 1 = -\frac{1}{2}x + \frac{3}{2}$$

$$\underline{y = -\frac{1}{2}x + \frac{5}{2}},$$

$$-\frac{1}{2}x + \frac{5}{2} = 0$$

$$\underline{x = 5},$$

$$\underline{(5, 0)},$$

$$1.3. \quad m_{AB} \times m_{BC}$$

$$= \frac{1+2}{3+3} \times \frac{-2-6}{-3+7}$$

$$= \frac{1}{2} \times \frac{-8}{4}$$

$$= \frac{1}{2} \times -2$$

$$= \underline{-1},$$

$$\therefore AB \perp BC$$

(2)

1.4

$$AB = \sqrt{(3+3)^2 + (1+2)^2}$$

$$= \underline{3\sqrt{5}}$$

$$BC = \sqrt{(-3+7)^2 + (-2-6)^2}$$

$$= \underline{4\sqrt{5}}$$

$$\begin{aligned} \text{Area } \triangle ABC &= \frac{1}{2} AB \cdot BC \\ &= \frac{1}{2} (3\sqrt{5})(4\sqrt{5}) \\ &= \underline{30 \text{ units}^2} \end{aligned}$$

1.5

$$m_{BD} \cdot m_{AC} = -1$$

$$m_{BD} \cdot -\frac{1}{2} = -1$$

$$m_{BD} = 2$$

$$y - y_1 = m(x - x_1)$$

$$y + 2 = 2(x + 3)$$

$$y + 2 = 2x + 6$$

$$y = \underline{2x + 4}$$

1.6

$$\alpha = 180^\circ - \tan^{-1}(2)$$

$$= \underline{116,57^\circ}$$

$$\beta = 180^\circ - \tan^{-1}\left(\frac{1}{2}\right)$$

$$= \underline{153,43^\circ}$$

$$\hat{BAC} = \beta - \alpha \quad \text{ext } \angle \text{ of } \triangle$$

$$= 153,43^\circ - 116,57^\circ$$

$$= 36,86^\circ$$

$$\approx \underline{37^\circ}$$

Question 2

2.1. $y = -3x + k$ sub $(-1, -3)$
 $-3 = -3(-1) + k$
 $-3 = 3 + k$
 $-6 = k$

2.2. $\frac{x_B + (-1)}{2} = 2,5$ $\frac{y_B - 3}{2} = 1$
 $x_B - 1 = 5$ $y_B - 3 = 2$
 $x_B = 6$ $y_B = 5$
B (6, 5)

2.3. $m_{AB} = \frac{1 - (-3)}{2,5 - (-1)}$
 $= \frac{4}{3,5}$
 $= \frac{8}{7}$

2.4. $\alpha = \tan^{-1}\left(\frac{8}{7}\right)$
 $= 48,81^\circ$

$\beta = 180^\circ - \tan^{-1}(3)$
 $= 108,43^\circ$

$\beta = \alpha + \theta$ ext $\angle$ of Δ
 $\theta = 108,43^\circ - 48,81^\circ$
 $= 59,62^\circ$

2.5. D (0, -6)

$AD = \sqrt{(-1-0)^2 + (-3+6)^2}$
 $= \sqrt{10}$

2.6

$$AC = 2\sqrt{10}$$

In ΔACB

$$BC^2 = AC^2 + AB^2 - 2AC \cdot AB \cos \theta$$

$$= (2\sqrt{10})^2 + (\sqrt{13})^2 - 2(2\sqrt{10})(\sqrt{13}) \cos 59.62^\circ$$

$$= 84.9982$$

$$BC = 9.22 \text{ units}$$

Question 3

$$3.1. M \left(\frac{9-4}{2} ; \frac{-2+2}{2} \right)$$

$$M \left(\frac{5}{2} ; 0 \right)$$

$$3.2. \quad m_{AC} = m_{BD} \quad \text{diag of parm}$$

$$\frac{x+8}{2} = \frac{5}{2}$$

$$x+8 = 5$$

$$\underline{x = -3}$$

$$\frac{y+6}{2} = 0$$

$$y+6 = 0$$

$$\underline{y = -6}$$

3.3

$$m_{BC} = \frac{-2+6}{9-8}$$

$$= \underline{4}$$

3.4

$$m_{AB} = \frac{-2-6}{9+3}$$

$$= -\frac{2}{3}$$

$$y - y_1 = m(x - x_1)$$

$$y + 2 = -\frac{2}{3}(x - 9)$$

$$y + 2 = -\frac{2}{3}x + 6$$

$$\underline{y = -\frac{2}{3}x + 4}$$

(5)

$$3.5 \quad \beta = 180^\circ - \tan^{-1}\left(\frac{2}{3}\right) \\ = 146,31^\circ$$

$$\alpha = \tan^{-1}(4) \\ = 75,96^\circ$$

$$\theta = \beta - \alpha \\ = 146,31^\circ - 75,96^\circ \\ = \underline{70,35^\circ}$$

$$3.6 \quad m_{BD} \times m_{BC}$$

$$= \frac{2+2}{-4-9} \times 4$$

$$= \frac{-16}{13} \neq -1$$

$\therefore \triangle DBC$ is not right angled.

Question 4

$$4.1 \quad A \left(\frac{3-1}{2}, \frac{5-7}{2} \right)$$

$$A = (1, -1)$$

$$4.2 \quad AP^2 = (3-1)^2 + (5+1)^2 \\ = 4 + 36 \\ = 40$$

$$\underline{(x-1)^2 + (y+1)^2 = 40}$$

Question 5:

5.1. $x^2 - 2x + y^2 + 4y = a$
 $x^2 - 2x + 1 + y^2 + 4y + 4 = a + 1 + 4$
 $(x-1)^2 + (y+2)^2 = a+5$

$$a+5 = r^2$$

$$a+5 = 5^2$$

$$a+5 = 25$$

$$\underline{a=20}$$

5.2. $M(1, -2)$

5.3. Sub $x=4$ into circle

$$4^2 - 2(4) + y^2 + 4y = 20$$

$$8 + y^2 + 4y = 20$$

$$y^2 + 4y - 12 = 0$$

$$(y+6)(y-2) = 0$$

$$y = -6$$

OR

$$\underline{y=2}$$

N/A

A(4, 2)

5.4 $m_{rad} = \frac{-2-2}{1-4} = \underline{\underline{\frac{4}{3}}}$

$$m_{tan} \times m_{rad} = -1$$

$$m_{tan} \times \frac{4}{3} = -1$$

$$m_{tan} = \underline{\underline{-\frac{3}{4}}}$$

$$y - y_1 = m(x - x_1)$$

$$y - 2 = -\frac{3}{4}(x - 4)$$

$$y - 2 = -\frac{3}{4}x + 3$$

$$y = -\frac{3}{4}x + 5$$

5.6. Centre: $(1 - 3, -2 + 1)$
 $(-2, -1)$

$$(x + 2)^2 + (y + 1)^2 = 25$$

Question 6

6.1. $r^2 = \left(\frac{5}{2} - 0\right)^2 + (2 - 0)^2$
 $= 4\frac{1}{4}$

$$(x - \frac{5}{2})^2 + (y - 2)^2 = 4\frac{1}{4}$$

$$x^2 - 5x + \frac{25}{4} + y^2 - 4y + 4 = 4\frac{1}{4}$$

$$x^2 + y^2 - 5x - 4y = 0$$

6.2. $m_{BC} = \frac{2 - 0}{\frac{5}{2} - 5}$
 $= -\frac{4}{5}$

$m_{LK} \cdot m_{BC} = -1$
 $m_{LK} = \frac{5}{4}$

tan $\perp$ radius

$$y - y_1 = m(x - x_1)$$

$$y - 0 = \frac{5}{4}(x - 5)$$

$$y = \frac{5}{4}x - \frac{25}{4}$$

6.3 $\angle AOB = 78,69^\circ$ tan chord theorem

6.4 $m = \tan 78,69^\circ$

$m = 5$,

$y = 5x$,

6.5 $y = 5x$ ① and $x^2 + y^2 - 5x - 4y = 0$ ②

① into ②

$$x^2 + (5x)^2 - 5x - 4(5x) = 0$$

$$26x^2 - 25x = 0$$

$$x(26x - 25) = 0$$

$$x = 0$$

$$x = \frac{25}{26}$$

N.A.

• 26

$$y = 5\left(\frac{25}{26}\right)$$

$$= \frac{125}{26}$$

26

$$A\left(\frac{25}{26}; \frac{125}{26}\right)$$

6.6 $R - r = \sqrt{6\frac{1}{4}} - \sqrt{4\frac{1}{4}} = 0,70$

$R + r = \sqrt{6\frac{1}{4}} + \sqrt{4\frac{1}{4}} = 7,10$

$$d^2 = \left(\frac{5}{2} + \frac{5}{2}\right)^2 + (2-3)^2$$

$$= 25 + 1$$

$$= 26$$

$$d = 5,10$$

$$\therefore R - r < d < R + r$$

$\therefore$ Two points of intersection